PAV_association_template

September 12, 2023

[1]:
```
# PAV association with BNI trait (Y~PAV)
# author: Sheng-Kai Hsu
# created: 2023.03.29
# last edited: 2023.09.12
# note: update to v3: papalum peptide query -> miniProt -> filter (80% CDS␣
 ↪coverage)
# repurposed for a template for panandGWAS script
```

[2]:
```
rm(list=ls())
library(limma)
library(scales)
library(pheatmap)
library(ape)
require(reshape2)
library(dplyr)
library(phytools)
library(geiger)
library(ggtree)
library(stringr)
library(qqman)

require(rMVP)      # R version of the MVP package
source("/workdir/sh2246/p_evolBNI/src/S01_phyloK.R")
```

```
Loading required package: reshape2


Attaching package: ‘dplyr’


The following object is masked from ‘package:ape’:

    where


The following objects are masked from ‘package:stats’:

    filter, lag
```

```
The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union

Loading required package: maps

Attaching package: ‘phytools’

The following object is masked from ‘package:scales’:

    rescale

ggtree v3.4.4 For help: https://yulab-smu.top/treedata-book/
If you use the ggtree package suite in published research, please cite
the appropriate paper(s):
Guangchuang Yu, David Smith, Huachen Zhu, Yi Guan, Tommy Tsan-Yuk Lam.
ggtree: an R package for visualization and annotation of phylogenetic
trees with their covariates and other associated data. Methods in
Ecology and Evolution. 2017, 8(1):28-36. doi:10.1111/2041-210X.12628
Guangchuang Yu. Using ggtree to visualize data on tree-like structures.
Current Protocols in Bioinformatics. 2020, 69:e96. doi:10.1002/cpbi.96
Shuangbin Xu, Lin Li, Xiao Luo, Meijun Chen, Wenli Tang, Li Zhan, Zehan
Dai, Tommy T. Lam, Yi Guan, Guangchuang Yu. Ggtree: A serialized data
object for visualization of a phylogenetic tree and annotation data.
iMeta 2022, 4(1):e56. doi:10.1002/imt2.56

Attaching package: ‘ggtree’

The following object is masked from ‘package:ape’:

    rotate
```

```
Loading required package: rMVP

Full description, Bug report, Suggestion and the latest version:

https://github.com/xiaolei-lab/rMVP
```

[3]:

```
GWAS=function(P,G,K,map,nPC=0,method="GLM",CV=NULL){
  # preparing the inputs (PAV, K matrix and map) according to MVP needs

  genotype_PAV     = as.big.matrix(G) # dimension: PAV x samples
  genotype_Kmatrix = as.big.matrix(K)          # dimension: samples x samples

  # perenniality  # data.frame, dimension: samples x (sample id + traits)
  # map_panand2   # data.frame, dimension: PAVs x 3 (gene, chr and position)

  #head(map_panand2)
  #map_panand2$chr = as.numeric(gsub(map_panand2$chr,pattern =␣
↪'chr',replacement = '')) # ignore it. does not matter

  GWAS_perenniality <- rMVP::MVP(
    # Data Inputs
    phe  = P,
    geno = genotype_PAV,
    K    = genotype_Kmatrix,
    map  = map,
    CV.GLM = CV, CV.MLM = CV,

    # Method
    method=method,

    # Number of Principal components for each mode ---
    nPC.GLM = nPC, nPC.MLM = nPC,

    # Threshold
    permutation.threshold = F, threshold = 0.05,

    # Outputs and computing parameters
    file.output=F, ncpus=8)
}
```

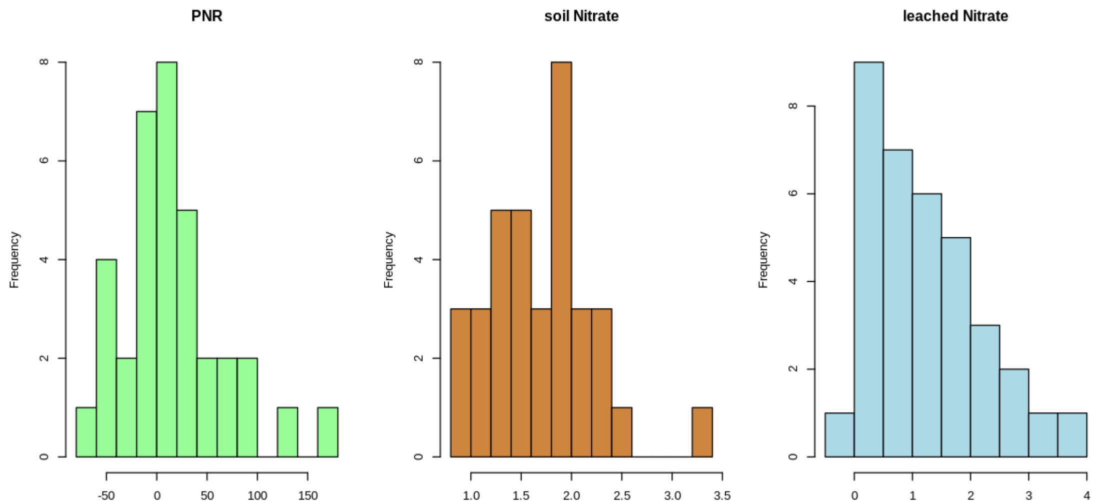
[4]:

```
# phenotype processing
dat=read.csv("/workdir/sh2246/p_evolBNI/data/Cornell_PNRandNitrateAssay.
↪csv",header = T)
annualSp=c("Zea mays","Sorghum bicolor","Thelepogon elegans")

#BNI calculation
```

```
dat$taxa=strsplit2(dat$plantID_repl.,"_")[,1]
control=median(dat$PNR[dat$taxa=="blank"])
dat$spTaxa=paste0(dat$taxa,": ",dat$species.name)
dat$BNI=(1-dat$PNR/control)*100

#Nitrate conversion
dat$soilN_mgN=dat$soilN*0.005/(0.002*(1-dat$SoilMoisture_dw*dat$DSW/dat$FSW))
dat$resinN_mgN=dat$resinN*0.005/0.002

dat_N_std=c(7.871,7.834,8.024,
            1.4,1.424,1.382,
            0.675,0.66,0.664)

#filtering
dat_filtered=dat[dat$inTree!=0&dat$condition!=0&!
 ↪dat$plantID%in%c("A10260002","A10280002","A10310002"),]
```

[5]:
```
se=function(x) sd(x)/sqrt(length(x))
BNI_avg=tapply(dat_filtered$BNI,dat_filtered$taxa,median)
BNI_se=tapply(dat_filtered$BNI,dat_filtered$taxa,se)

soilN_avg=tapply(dat_filtered$soilN_mgN,dat_filtered$taxa,median)
soilN_se=tapply(dat_filtered$soilN_mgN,dat_filtered$taxa,se)

resinN_avg=tapply(dat_filtered$resinN_mgN,dat_filtered$taxa,median)
resinN_se=tapply(dat_filtered$resinN_mgN,dat_filtered$taxa,se)

pwN_T0_avg=tapply(dat_filtered$T0,dat_filtered$taxa,median)
pwN_T0_se=tapply(dat_filtered$T0,dat_filtered$taxa,se)
pwN_T1_avg=tapply(dat_filtered$T1,dat_filtered$taxa,median)
pwN_T1_se=tapply(dat_filtered$T1,dat_filtered$taxa,se)
pwN_T2_avg=tapply(dat_filtered$T2,dat_filtered$taxa,median)
pwN_T2_se=tapply(dat_filtered$T2,dat_filtered$taxa,se)
pwN_T3_avg=tapply(dat_filtered$T3,dat_filtered$taxa,median)
pwN_T3_se=tapply(dat_filtered$T3,dat_filtered$taxa,se)
```

[6]:
```
options(repr.plot.width=10, repr.plot.height=5)
par(mfrow=c(1,3))
hist(-BNI_avg,breaks = 10,main = "PNR",xlab = "",col="palegreen")
hist(log(soilN_avg+0.01),breaks = 10,main = "soil Nitrate",xlab = "",col =␣
 ↪"tan3")
hist(log(resinN_avg+0.01),breaks = 10,main = "leached Nitrate",xlab = "",col =␣
 ↪"lightblue")
```



[7]:
```
#phenotype matrix generation
pheno=data.frame(sample=names(BNI_avg),PNR=-BNI_avg,soilN␣
  ↪=soilN_avg,resinN=resinN_avg)
pheno1=data.frame(sample=names(BNI_avg),pheno=BNI_avg)
pheno2=data.frame(sample=names(soilN_avg),pheno=log(soilN_avg+0.01))
pheno3=data.frame(sample=names(resinN_avg),pheno=log(resinN_avg+0.01))
```

[8]:
```
metadata=read.csv("/workdir/sh2246/p_evolBNI/data/BNITaxa_translateTable.
  ↪csv",header = T)
```

[9]:
```
#load genomic PAV data
pav=read.table("/workdir/sh2246/p_evolBNI/output/PAV_CNV/Pv_miniProt_PAV.
  ↪txt",header = T)
```

[10]:
```
#name matching
merge_tab=data.frame(Sample=strsplit2(gsub("X","",colnames(pav)),split = "_|.
  ↪final")[,1])
metadata$Sample=gsub("-",".",strsplit2(metadata$genome_path,split = "_|.final|.
  ↪fa")[,1])
metadata$spTaxa=paste0(metadata$Plant_ID,": ",metadata$species.name)
merge_tab=merge(merge_tab,metadata,by = "Sample")
merge_tab=merge_tab[!duplicated(merge_tab$Sample),]
rownames(merge_tab)=merge_tab$Sample
merge_tab=merge_tab[strsplit2(gsub("X","",colnames(pav)),split = "_|.
  ↪final")[,1],]

colnames(pav)=merge_tab$Plant_ID
```

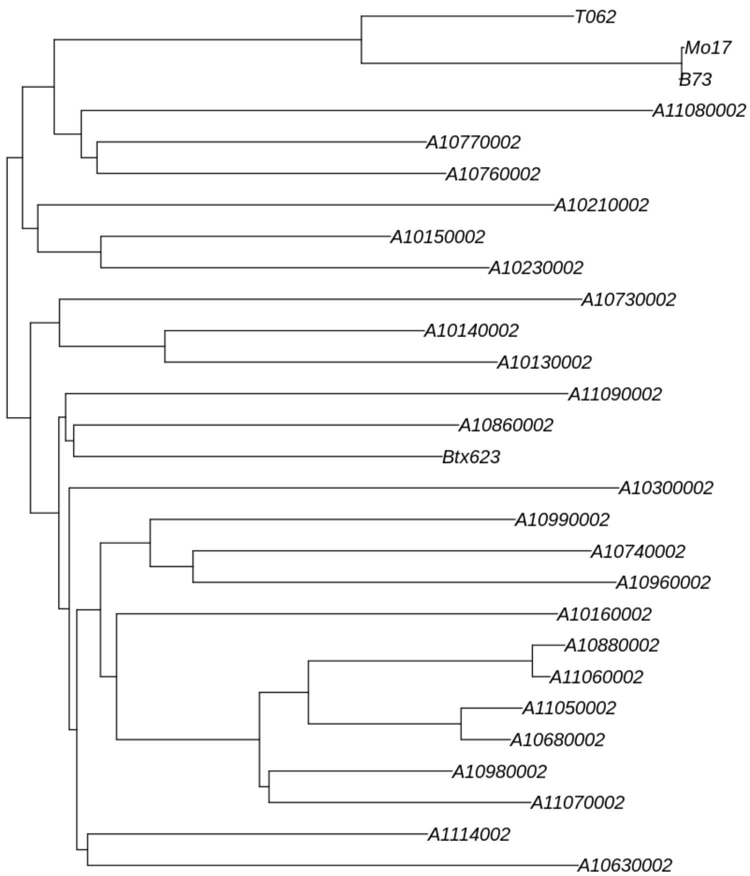
[11]:
```
# neutral tree input and renaming
rootedtree1 = read.tree("/workdir/sh2246/p_evolBNI/output/
↪speciesTrees_median4dD_v2.newick")

rootedtree1.renamed=rootedtree1
rootedtree1.renamed$tip.label = gsub("-",".",rootedtree1.renamed$tip.label)
rootedtree1.renamed$tip.label = merge_tab[rootedtree1.renamed$tip.
↪label,]$Plant_ID
rootedtree1.renamed$tip.label
```

1. 'A10630002' 2. 'A1114002' 3. 'A11070002' 4. 'A10980002' 5. 'A10680002' 6. 'A11050002' 7. 'A11060002' 8. 'A10880002' 9. 'A10250002' 10. 'A11100002' 11. 'A10160002' 12. 'A10870002' 13. 'A10960002' 14. 'A10740002' 15. 'A10100002' 16. 'A10200002' 17. 'A10990002' 18. 'A10300002' 19. 'Btx623' 20. 'A10330002' 21. 'A10860002' 22. 'A11090002' 23. 'A10130002' 24. 'A10140002' 25. 'A10730002' 26. 'A10230002' 27. 'A10150002' 28. 'A10210002' 29. 'A10760002' 30. 'A10770002' 31. 'A11080002' 32. 'B73' 33. 'Mo17' 34. 'T062' 35. NA

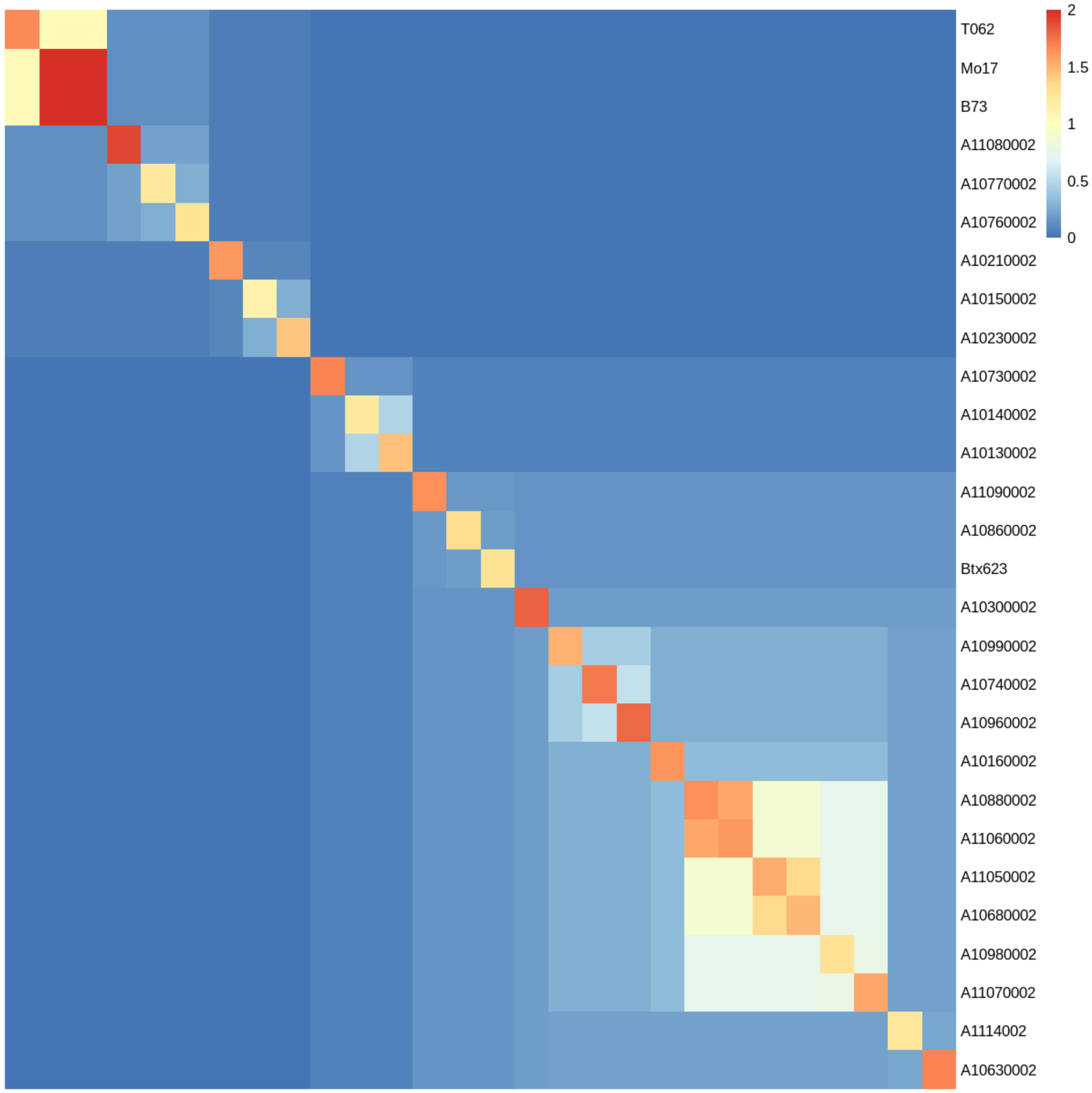
[12]:
```
#load gff for position mapping
gff=read.delim("/workdir/sh2246/p_evolBNI/data/Pvaginatum_672_v3.1.
↪gene_exons_modified_sorted2.gff3",header = F)
gff_genePos=gff[gff$V3=="mRNA",]
rownames(gff_genePos)=gsub("ID=","",strsplit2(gff_genePos$V9,";")[,1])
```

[13]:
```
#position mapping and PAV filtering
map = data.frame(gene=rownames(gff_genePos),chr =␣
↪gff_genePos[,1],POS=gff_genePos[,4])
map=map[grep("Chr",map$chr),]
map$gene = as.character(map$gene)
map = map [map$gene %in% rownames(pav),]

PAVUsed=pav[map$gene,colnames(pav)%in% rootedtree1.renamed$tip.label]
PAVUsed=PAVUsed[,colSums(PAVUsed)>30000]
PAVUsed=PAVUsed[rowSums(PAVUsed)>1&rowSums(PAVUsed)<(33-6),]

map2=map[map$gene%in%rownames(PAVUsed),]
```

[14]:
```
options(repr.plot.width=10, repr.plot.height=10)
rootedtree1.renamed$tip.label[35] = "ref"
inputTree=keep.tip(rootedtree1.renamed,colnames(PAVUsed))
plot(inputTree,use.edge.length = T)
```



[15]:
```
#phyloK calculation
K=phyloK(inputTree )
colnames(K)=rownames(K)
pheatmap(K[rev(inputTree$tip.label),rev(inputTree$tip.label)],cluster_cols =␣
↪F,cluster_rows = F,
         show_rownames = T,show_colnames = F,border_color = NA)
```



[16]:
```
# run GWAS
res=GWAS(pheno1[colnames(PAVUsed[,inputTree$tip.
  ↪label]),],PAVUsed[,inputTree$tip.label],K,map2,method = c("GLM","MLM"))
res2=GWAS(pheno2[colnames(PAVUsed[,inputTree$tip.
  ↪label]),],PAVUsed[,inputTree$tip.label],K,map2,method = c("GLM","MLM"))
res3=GWAS(pheno3[colnames(PAVUsed[,inputTree$tip.
  ↪label]),],PAVUsed[,inputTree$tip.label],K,map2,method = c("GLM","MLM"))
```

```
Warning message in as.big.matrix(G):
"Coercing data.frame to matrix via factor level numberings."
```

```
====================== Welcome to MVP ======================
      A Memory-efficient, Visualization-enhanced, and
             Parallel-accelerated Tool For GWAS
```

```
                 __  __  __   __  ___
                |  \/  | \ \ / / | _ \
                | |\/| |  \ V /  |  _/
                |_|  |_|   \_/   |_|     Version: 1.0.6
  Designed and Maintained by Lilin Yin, Haohao Zhang, and
  Xiaolei Liu
  Contributors: Zhenshuang Tang, Jingya Xu, Dong Yin,
  Zhiwu Zhang, Xiaohui Yuan, Mengjin Zhu, Shuhong Zhao,
  Xinyun Li
  Contact: xiaoleiliu@mail.hzau.edu.cn
============================================================
Start: 2023-09-12 10:19:50
Input data has 28 individuals, 8415 markers
Analyzed trait: pheno
Number of threads used: 8
Total 2 individuals removed due to missings
Eigen Decomposition on GRM
-------------------------GWAS Start-------------------------
General Linear Model (GLM) Start…
scanning…
Genomic inflation factor (lambda): 1.2317
Mixed Linear Model (MLM) Start…
Variance components using: BRENT
Estimated Vg and Ve: 2334.936836 2.962172
scanning…
Genomic inflation factor (lambda): 1.7996
Significant level: 5.94e-06
End: 2023-09-12 10:19:52
===================== MVP ACCOMPLISHED =====================

Warning message in as.big.matrix(G):
"Coercing data.frame to matrix via factor level numberings."

====================== Welcome to MVP ======================
      A Memory-efficient, Visualization-enhanced, and
             Parallel-accelerated Tool For GWAS

                 __  __  __   __  ___
                |  \/  | \ \ / / | _ \
                | |\/| |  \ V /  |  _/
                |_|  |_|   \_/   |_|     Version: 1.0.6
  Designed and Maintained by Lilin Yin, Haohao Zhang, and
  Xiaolei Liu
  Contributors: Zhenshuang Tang, Jingya Xu, Dong Yin,
  Zhiwu Zhang, Xiaohui Yuan, Mengjin Zhu, Shuhong Zhao,
  Xinyun Li
  Contact: xiaoleiliu@mail.hzau.edu.cn
============================================================
Start: 2023-09-12 10:19:53
Input data has 28 individuals, 8415 markers
```

```
Analyzed trait: pheno
Number of threads used: 8
Total 2 individuals removed due to missings
Eigen Decomposition on GRM
-------------------------GWAS Start-------------------------
General Linear Model (GLM) Start…
scanning…
Genomic inflation factor (lambda): 0.8621
Mixed Linear Model (MLM) Start…
Variance components using: BRENT
Estimated Vg and Ve: 0.190951 0.055816
scanning…
Genomic inflation factor (lambda): 0.4657
Significant level: 5.94e-06
End: 2023-09-12 10:19:55
==================== MVP ACCOMPLISHED ====================

Warning message in as.big.matrix(G):
"Coercing data.frame to matrix via factor level numberings."

===================== Welcome to MVP =====================
      A Memory-efficient, Visualization-enhanced, and
             Parallel-accelerated Tool For GWAS

                 __  __  __   __  ___
                |  \/  | \ \ / / | _ \
                | |\/| |  \ V /  |  _/
                |_|  |_|   \_/   |_|     Version: 1.0.6
  Designed and Maintained by Lilin Yin, Haohao Zhang, and
  Xiaolei Liu
  Contributors: Zhenshuang Tang, Jingya Xu, Dong Yin,
  Zhiwu Zhang, Xiaohui Yuan, Mengjin Zhu, Shuhong Zhao,
  Xinyun Li
  Contact: xiaoleiliu@mail.hzau.edu.cn
==========================================================
Start: 2023-09-12 10:19:56
Input data has 28 individuals, 8415 markers
Analyzed trait: pheno
Number of threads used: 8
Total 2 individuals removed due to missings
Eigen Decomposition on GRM
-------------------------GWAS Start-------------------------
General Linear Model (GLM) Start…
scanning…
Genomic inflation factor (lambda): 0.6134
Mixed Linear Model (MLM) Start…
Variance components using: BRENT
Estimated Vg and Ve: 0.893538 0.000405
scanning…
Genomic inflation factor (lambda): 1.9792
```

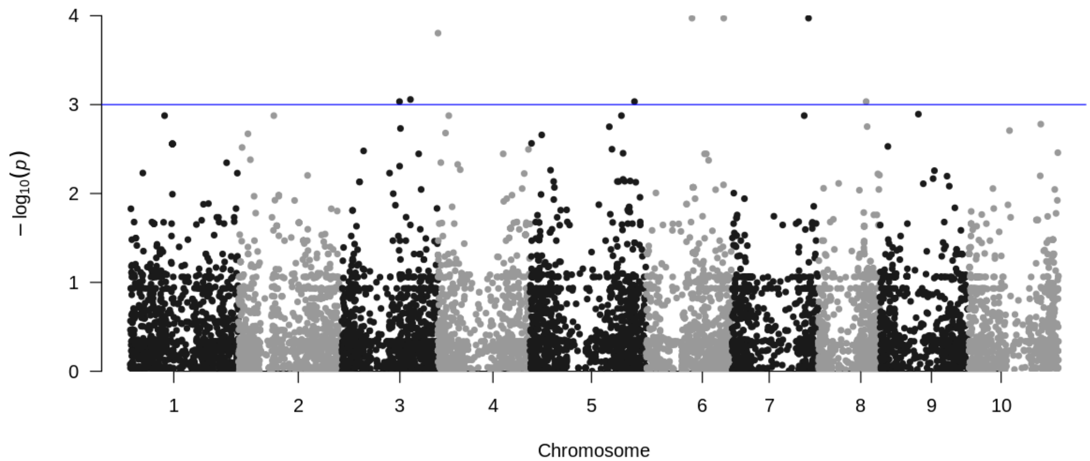
```
Significant level: 5.94e-06
End: 2023-09-12 10:19:58
===================== MVP ACCOMPLISHED =====================
```

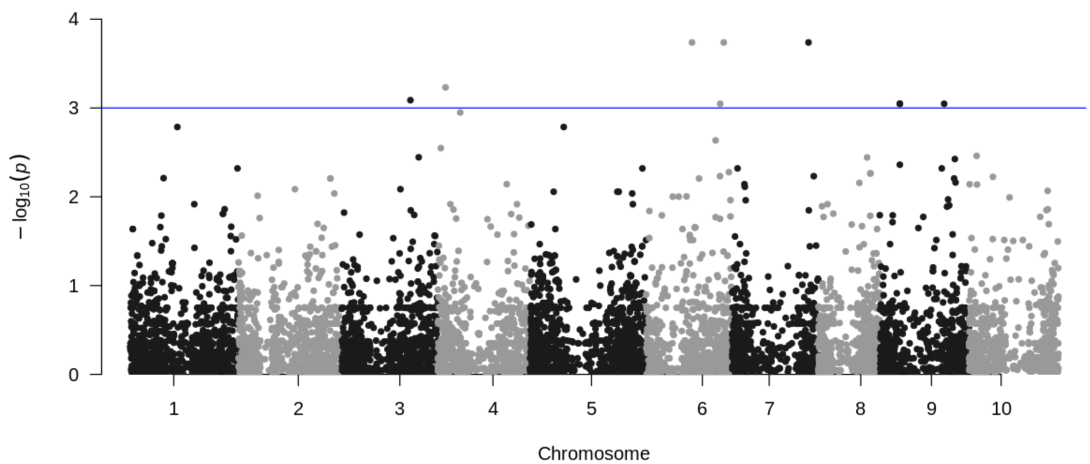
[17]:
```
head(res$glm.results)
```

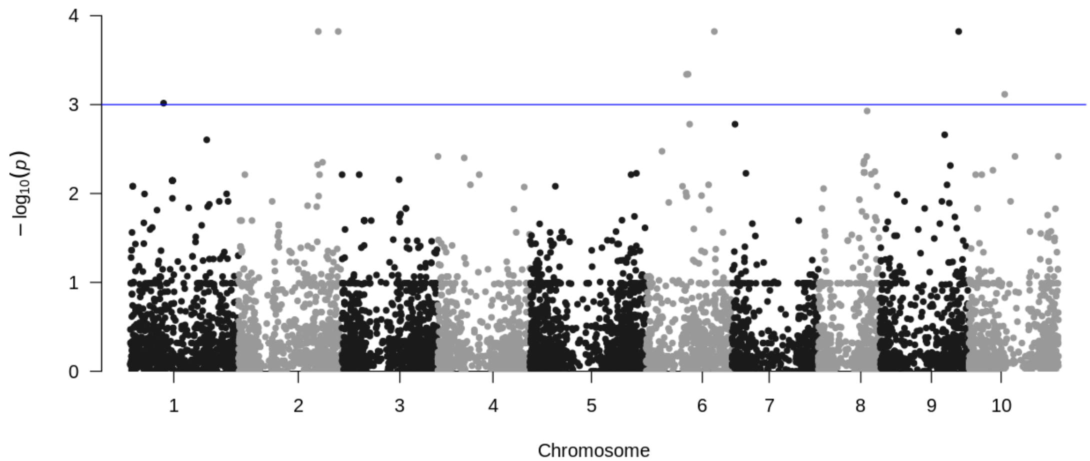
A matrix: 6 × 3 of type dbl

Effect	SE	pheno.GLM
-5.894384	28.25393	0.83650403
-26.843372	27.74363	0.34291854
4.403579	41.81635	0.91700695
-25.701298	22.83228	0.27144208
-64.545776	39.69666	0.11701568
-51.721900	19.70706	0.01485558

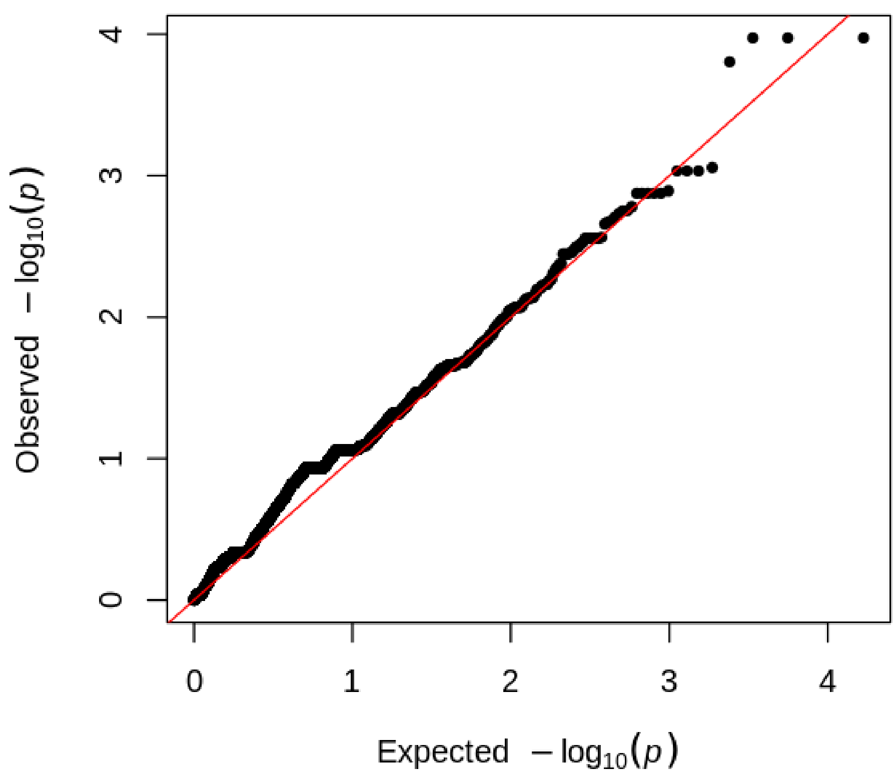
[19]:
```
# manhattan and qq plot
res_glm=data.frame(Chr=as.numeric(gsub("Chr","",map2$chr)),BP=map2$POS,P_PNR=as.
↪numeric(res$glm.results[,3]),
                   P_soil=as.numeric(res2$glm.results[,3]),P_resin=as.
↪numeric(res3$glm.results[,3]), Gene=map2$gene)
res_mlm=data.frame(Chr=as.numeric(gsub("Chr","",map2$chr)),BP=map2$POS,P_PNR=as.
↪numeric(res$mlm.results[,3]),
                   P_soil=as.numeric(res2$mlm.results[,3]),P_resin=as.
↪numeric(res3$mlm.results[,3]), Gene=map2$gene)

options(repr.plot.width=10, repr.plot.height=5)
manhattan(na.omit(res_glm),chr = "Chr",bp = "BP",p = "P_PNR",snp = "Gene",logp␣
↪= T,suggestiveline = 3,genomewideline = F)
# manhattan(na.omit(res_mlm),chr = "Chr",bp = "BP",p = "P_PNR",snp =␣
↪"Gene",logp = T,suggestiveline = 3,genomewideline = F)
manhattan(na.omit(res_glm),chr = "Chr",bp = "BP",p = "P_soil",snp = "Gene",logp␣
↪= T,suggestiveline = 3,genomewideline = F)
# manhattan(na.omit(res_mlm),chr = "Chr",bp = "BP",p = "P_soil",snp =␣
↪"Gene",logp = T,suggestiveline = 3,genomewideline = F)
manhattan(na.omit(res_glm),chr = "Chr",bp = "BP",p = "P_resin",snp =␣
↪"Gene",logp = T,suggestiveline = 3,genomewideline = F)
# manhattan(na.omit(res_mlm),chr = "Chr",bp = "BP",p = "P_resin",snp =␣
↪"Gene",logp = T,suggestiveline = 3,genomewideline = F)
options(repr.plot.width=5, repr.plot.height=5)
qq(as.numeric(res$glm.results[,3]))
qq(as.numeric(res$mlm.results[,3]))
qq(as.numeric(res2$glm.results[,3]))
qq(as.numeric(res2$mlm.results[,3]))
qq(as.numeric(res3$glm.results[,3]))
qq(as.numeric(res3$mlm.results[,3]))
```


4
3
2
1
0
$-\log_{10}(p)$
1
2
3
4
5
6
7
8
9
10
Chromosome

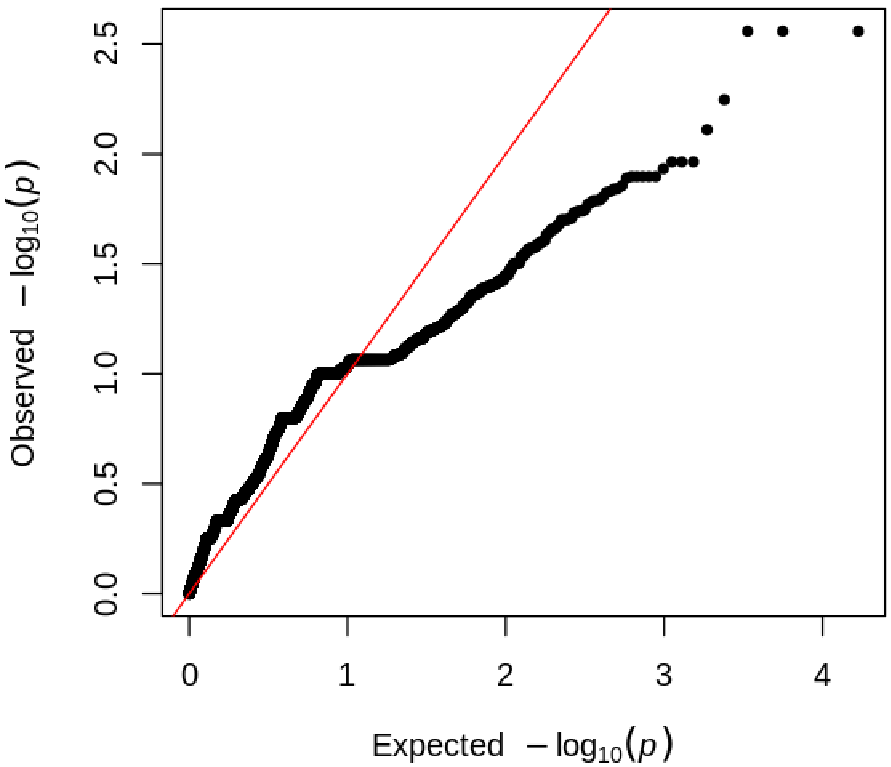

4
3
2
1
0
$-\log_{10}(p)$
1
2
3
4
5
6
7
8
9
10
Chromosome



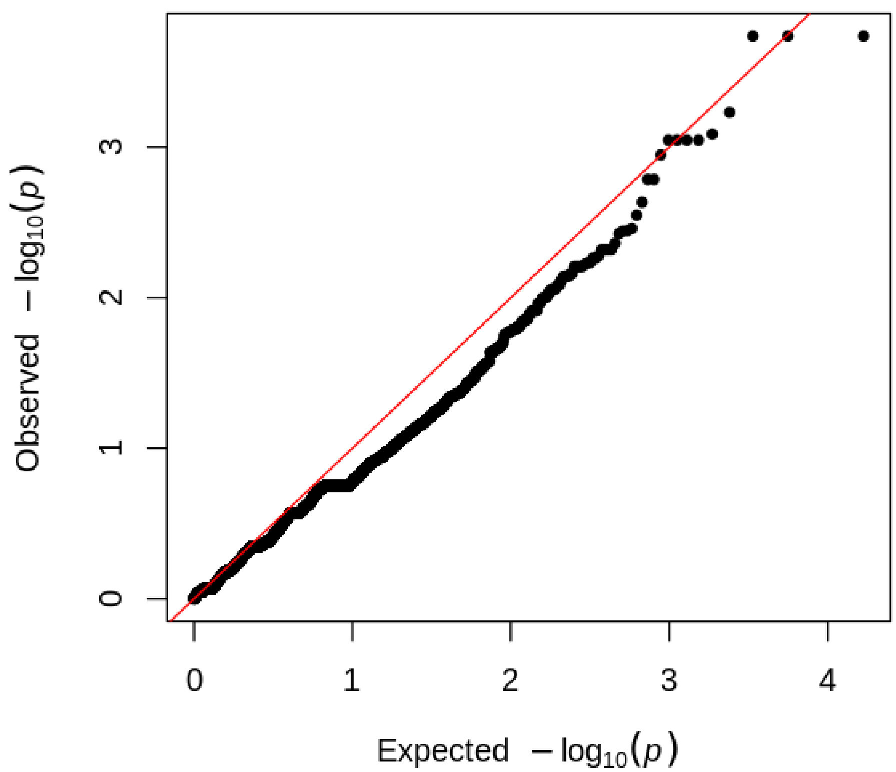
4
3
2
1
0
$-\log_{10}(p)$
1
2
3
4
5
6
7
8
9
10
Chromosome

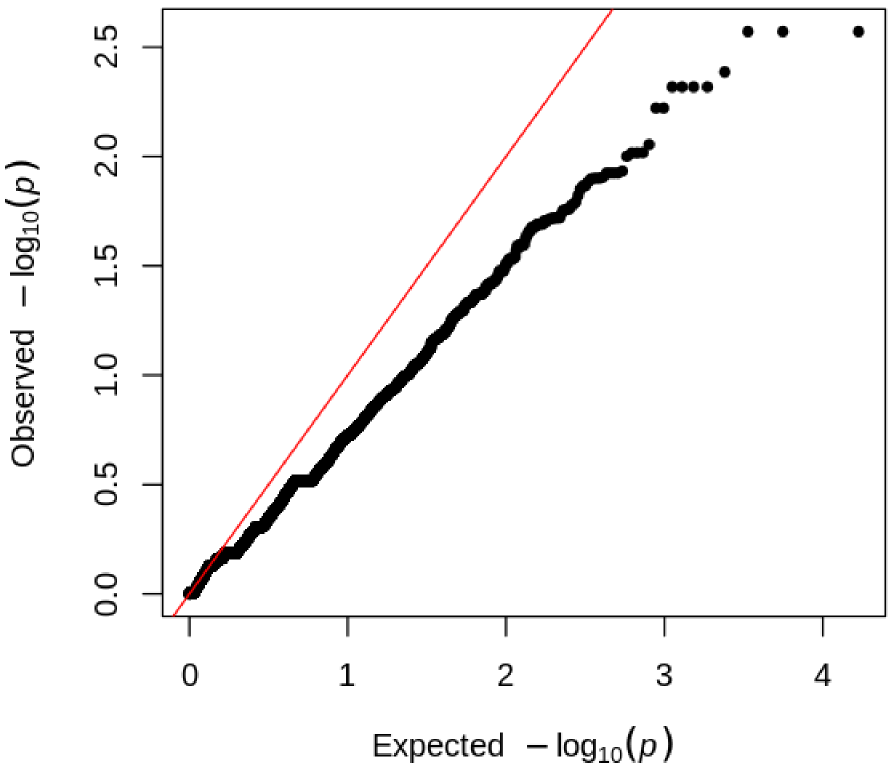


4
3
2
1
0
Observed $-\log_{10}(p)$
0
1
2
3
4
Expected $-\log_{10}(p)$

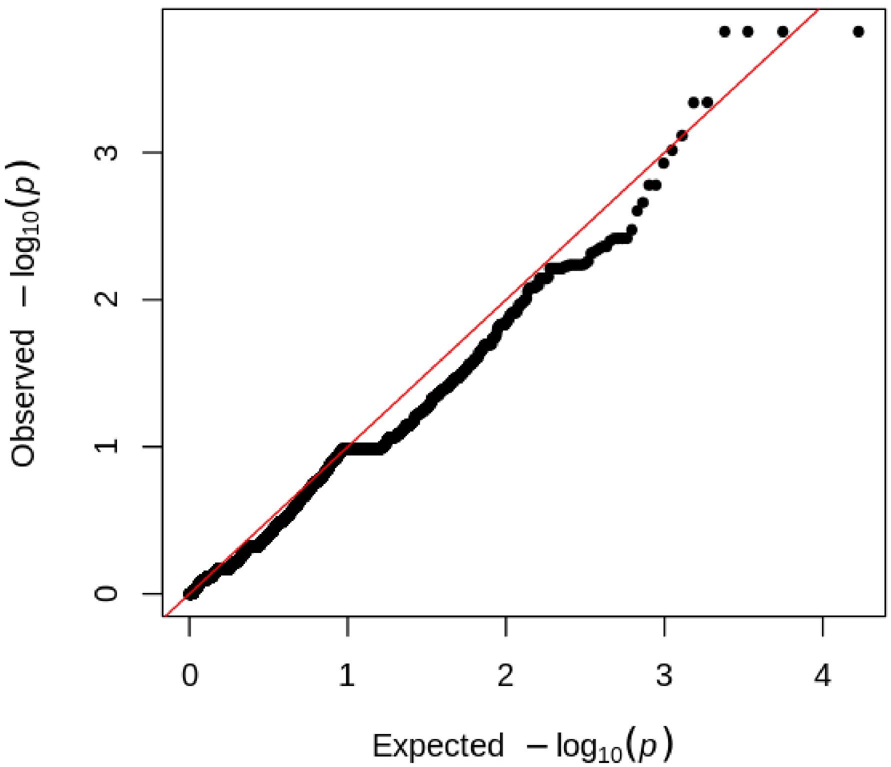


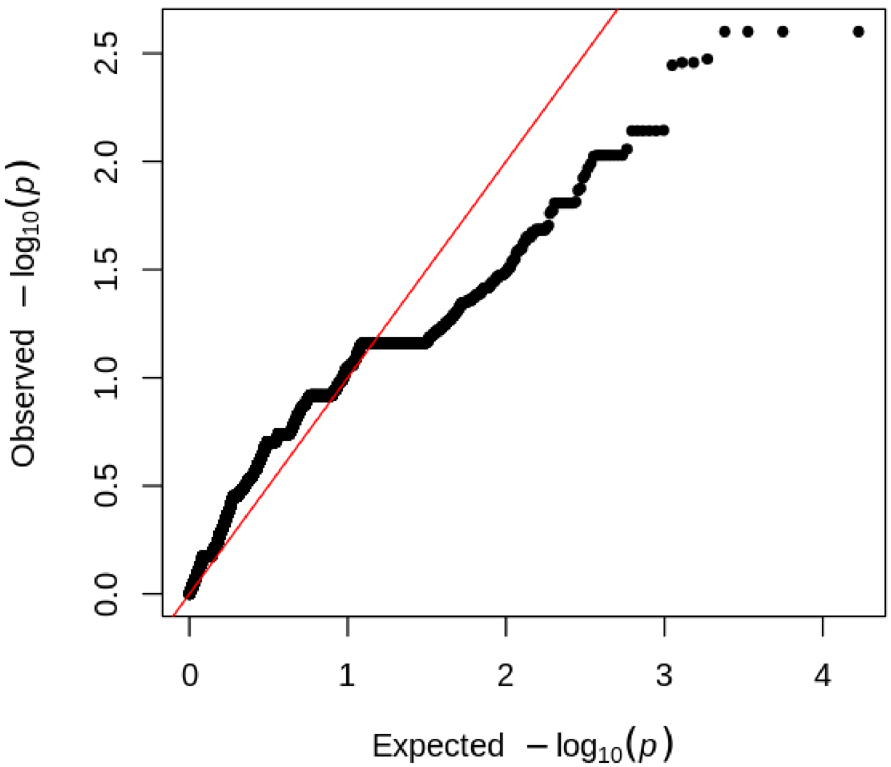
Observed $-\log_{10}(p)$
Expected $-\log_{10}(p)$
0.0
0.5
1.0
1.5
2.0
2.5
0
1
2
3
4


Observed $-\log_{10}(p)$
Expected $-\log_{10}(p)$
0
1
2
3
4



2.5
2.0
1.5
1.0
0.5
0.0
Observed $-\log_{10}(p)$
0
1
2
3
4
Expected $-\log_{10}(p)$


Observed $-\log_{10}(p)$
Expected $-\log_{10}(p)$
0
1
2
3
4



Observed $-\log_{10}(p)$
Expected $-\log_{10}(p)$
0.0
0.5
1.0
1.5
2.0
2.5
0
1
2
3
4

[ ]: